

```
USE JobMarketAnalysis;
GO
```

```
-- Vista rápida de los datos
```

```
Select COUNT(*) AS totalVacantes,
        COUNT(DISTINCT pais_origen) AS totalPaises,
        MIN(date_posted) AS fecha_mas_antigua,
        MAX(date_posted) AS fecha_mas_reciente
FROM vacantes_import
```

totalVacantes	totalPaises	fecha_mas_antigua	fecha_mas_reciente
2629	11	2022-10-13	2026-05-04

```
-- Ver cuántas vacantes tienen URL disponible
```

```
SELECT
    COUNT(*) as Total,
    SUM(CASE WHEN job_url IS NOT NULL THEN 1 ELSE 0 END) as ConJobUrl,
    SUM(CASE WHEN job_url_direct IS NOT NULL THEN 1 ELSE 0 END) as ConJobUrlDirect
FROM vacantes_import;
```

Total	ConJobUrl	ConJobUrlDirect
2629	2629	1636

--Conseguir Vacantes recientes (a partir del 2024) y contemplar las vacantes remotas por pais.

```
SELECT
    pais_origen,
    COUNT(*) as VacantesRemotas
FROM vacantes_import
WHERE
    -- Español
    LOWER(description) LIKE '%remoto%'
    OR LOWER(description) LIKE '%teletrabajo%'
    OR LOWER(description) LIKE '%desde casa%'
    OR LOWER(description) LIKE '%home office%'
    -- Inglés
    OR LOWER(description) LIKE '%remote%'
    OR LOWER(description) LIKE '%work from home%'
    OR LOWER(description) LIKE '%wfh%'
    OR LOWER(description) LIKE '%anywhere%'
    OR LOWER(description) LIKE '%virtual%'
    OR LOWER(description) LIKE '%hybrid%'
    OR LOWER(description) LIKE '%hibrido%'
    OR LOWER(description) LIKE '%flexible%'
    -- En el título
    OR LOWER(title) LIKE '%remoto%'
    OR LOWER(title) LIKE '%remote%'
    OR LOWER(title) LIKE '%home%'
GROUP BY pais_origen
ORDER BY VacantesRemotas DESC;
```

pais_origen	VacantesRemotas
USA	415
Canada	165
Brazil	115
Mexico	113
Spain	49
Colombia	31
Argentina	22
Chile	12
Costa Rica	9
Peru	5
Uruguay	4

--Investigaremos sobre las Skills mas demandadas

```
WITH SkillsAnalysis AS (
    SELECT
        CASE
            WHEN description LIKE '%SQL%' OR description LIKE '%sql%' THEN 'SQL'
            WHEN description LIKE '%Python%' OR description LIKE '%python%' THEN 'Python'
            WHEN description LIKE '%Power BI%' OR description LIKE '%power bi%' THEN 'Power BI'
            WHEN description LIKE '%Tableau%' OR description LIKE '%tableau%' THEN 'Tableau'
            WHEN description LIKE '%Excel%' OR description LIKE '%excel%' THEN 'Excel'
            WHEN description LIKE '%Pandas%' OR description LIKE '%pandas%' THEN 'Pandas'
            WHEN description LIKE '%ETL%' OR description LIKE '%etl%' THEN 'ETL'
            WHEN description LIKE '%DAX%' OR description LIKE '%dax%' THEN 'DAX'
        END as Skill
    FROM vacantes_import
    WHERE description IS NOT NULL
)
SELECT
    Skill,
    COUNT(*) as Menciones
FROM SkillsAnalysis
WHERE Skill IS NOT NULL
GROUP BY Skill
ORDER BY Menciones DESC;
```

Skill	Menciones
Excel	682
SQL	413
Power BI	124
Python	62
Tableau	22
ETL	12

--Vacantes en el nicho de experiencia

```
SELECT
    pais_origen,
    COUNT(*) as VacantesNicho
FROM vacantes_import
WHERE
    is_remote = 'TRUE' AND
    (description LIKE '%food%' OR description LIKE '%aliment%')
    OR (description LIKE '%supply%' OR description LIKE '%logistic%')
    OR (title LIKE '%food%' OR title LIKE '%supply%')

GROUP BY pais_origen
ORDER BY VacantesNicho DESC;
```

pais_origen	VacantesNicho
USA	328
Canada	72
Brazil	61
Mexico	33
Spain	14
Colombia	7
Argentina	5
Chile	2
Costa Rica	2
Peru	2
Uruguay	1

–Vacantes que piden inglés

```

SELECT
    pais_origen,
    COUNT(*) as VacantesConIngles
FROM vacantes_import
WHERE
    description LIKE '%english%'
    OR description LIKE '%inglés%'
    OR description LIKE '%bilingual%'
    OR title LIKE '%english%'
GROUP BY pais_origen
ORDER BY VacantesConIngles DESC;
```

pais_origen	VacantesConIngles
Mexico	74
USA	63
Brazil	62
Canada	52
Spain	47
Colombia	26
Argentina	18
Costa Rica	8
Chile	5
Uruguay	3

Peru	2
------	---

```

SELECT TOP 20
    company,
    pais_origen,
    COUNT(*) as Vacantes
FROM vacantes_import
WHERE company IS NOT NULL AND company != ""
GROUP BY company, pais_origen
ORDER BY Vacantes DESC;

```

company	pais_origen	Vacantes
Prolific	Mexico	68
Prolific	Colombia	31
Bunge	Brazil	15
Crossing Hurdles	Mexico	13
DiDi Global	Mexico	11
Amazon.com	USA	11
Adimax Industria e Comercio de Alimentos Ltda	Brazil	9
Eurofins	USA	9
KBR	USA	9
Lightspeed	Canada	8
Prolific	Canada	8
NBCUniversal	USA	8

Prolific	USA	8
Mercor	Mexico	7
Openkyber	USA	7
Prolific	Argentina	6
DiDi Global	Brazil	6
Jobbol	Brazil	6
Recutify Inc.	Canada	6
Jobs via Dice	Mexico	6

Prolific destaca como una empresa de posicionamiento global en búsqueda de perfiles de data analyst junior en toda america, ofreciendo condiciones de trabajo remoto

-- Vacantes por mes (últimos 12 meses)

```
SELECT
    FORMAT(date_posted, 'yyyy-MM') as Mes,
    COUNT(*) as TotalVacantes
FROM vacantes_import
WHERE date_posted >= DATEADD(MONTH, -12, GETDATE())
GROUP BY FORMAT(date_posted, 'yyyy-MM')
ORDER BY Mes desc;
```

Mes	TotalVacantes
2026-05	423
2026-04	2095
2026-03	12
2026-02	8
2026-01	4
2025-12	5
2025-10	2
2025-09	2

El total de vacantes por mes indica, no necesariamente que Abril es el mes con vacantes, sino que es el mes con la muestra mas grande, debido a que es el mas reciente de los meses completos, debido a que mayo esta en curso.

-- 2.1 Top skills por país (México, Colombia, Argentina, USA)

```
SELECT
    pais_origen,
    SUM(CASE WHEN LOWER(description) LIKE '%sql%' THEN 1 ELSE 0 END) as SQL,
    SUM(CASE WHEN LOWER(description) LIKE '%python%' THEN 1 ELSE 0 END) as Python,
    SUM(CASE WHEN LOWER(description) LIKE '%power bi%' THEN 1 ELSE 0 END) as PowerBI,
    SUM(CASE WHEN LOWER(description) LIKE '%excel%' THEN 1 ELSE 0 END) as Excel,
    SUM(CASE WHEN LOWER(description) LIKE '%tableau%' THEN 1 ELSE 0 END) as Tableau,
    COUNT(*) as TotalVacantes
FROM vacantes_import
WHERE pais_origen IN ('Mexico', 'Colombia', 'Argentina', 'USA', 'Spain')
GROUP BY pais_origen
ORDER BY TotalVacantes DESC;

USE JobMarketAnalysis;
GO
```

pais_origen	SQL	Python	PowerBI	Excel	Tableau	TotalVacantes
Argentina	5	3	2	12	2	75
Colombia	17	12	8	35	1	204
Mexico	27	13	30	106	13	692
Spain	27	18	14	48	11	91
USA	170	111	99	510	103	736

A pesar de la existencia de nuevas aplicaciones y tecnologías, de la practicidad y la utilidad de las herramientas de visualizacion, los skills mas demandados dentro del area de DA para juniors sigue siendo EXCEL Y SQL. Siendo TABLEAU de las herramientas menos demandadas

1.1 Porcentaje de vacantes que mencionan IA

```
SELECT
COUNT(*) as TotalVacantes,
SUM(CASE
  WHEN LOWER(description) LIKE '%ai%'
  OR LOWER(description) LIKE '%artificial intelligence%'
  OR LOWER(description) LIKE '%machine learning%'
  OR LOWER(description) LIKE '%genai%'
  OR LOWER(description) LIKE '%llm%'
  THEN 1 ELSE 0 END) as VacantesConIA,
ROUND(100.0 * SUM(CASE
  WHEN LOWER(description) LIKE '%ai%'
  OR LOWER(description) LIKE '%artificial intelligence%'
  OR LOWER(description) LIKE '%machine learning%'
  OR LOWER(description) LIKE '%genai%'
  OR LOWER(description) LIKE '%llm%'
  THEN 1 ELSE 0 END) / COUNT(*), 2) as PorcentajelA
FROM vacantes_import;
```

HabilidadIA	Menciones
Machine Learning / AI General	1522
RAG (Retrieval-Augmented Generation)	809
AI Agents / Agentic AI	98
LLM / Large Language Models	95
Prompt Engineering	46
Generative AI (GenAI)	35
MLOps / LLMOps	10

Son demandados perfiles con AI generativa y ML en mas de la mitad de los perfiles. Lo cual nos obliga a colocarlo SI o SI dentro de nuestras SKILLS para ser tomados en cuenta

2.1 Desglose por tipo de habilidad IA

```
SELECT
    'Machine Learning / AI General' as HabilidadIA,
    COUNT(*) as Menciones
FROM vacantes_import
WHERE LOWER(description) LIKE '%machine learning%'
    OR LOWER(description) LIKE '%artificial intelligence%'
    OR LOWER(description) LIKE '%ai %'
    OR LOWER(description) LIKE '% %ai%'
```

UNION ALL

```
SELECT
    'Generative AI (GenAI)',
    COUNT(*)
FROM vacantes_import
WHERE LOWER(description) LIKE '%generative ai%'
    OR LOWER(description) LIKE '%genai%'
    OR LOWER(description) LIKE '%gen ai%'
```

UNION ALL

```
SELECT
    'LLM / Large Language Models',
    COUNT(*)
FROM vacantes_import
WHERE LOWER(description) LIKE '%llm%'
    OR LOWER(description) LIKE '%large language model%'
```

UNION ALL

```
SELECT
    'Prompt Engineering',
    COUNT(*)
FROM vacantes_import
WHERE LOWER(description) LIKE '%prompt%'
    OR LOWER(description) LIKE '%prompt engineering%'
```

UNION ALL

```
SELECT
    'RAG (Retrieval-Augmented Generation)',
    COUNT(*)
FROM vacantes_import
WHERE LOWER(description) LIKE '%rag%'
    OR LOWER(description) LIKE '%retrieval%'
```

UNION ALL

```
SELECT
    'AI Agents / Agentic AI',
    COUNT(*)
FROM vacantes_import
WHERE LOWER(description) LIKE '%agent%'
    AND LOWER(description) LIKE '%ai%'
```

UNION ALL

```

SELECT
  'MLOps / LLMOps',
  COUNT(*)
FROM vacantes_import
WHERE LOWER(description) LIKE '%mlops%'
  OR LOWER(description) LIKE '%llmops%'
  OR LOWER(description) LIKE '%model deployment%'

ORDER BY Menciones DESC;

```

pais_origen	TotalVacantes	VacantesConIA	PorcentajeIA
USA	736	709	96,33%
Canada	292	265	90,75%
Spain	91	62	68,13%
Argentina	75	22	29,33%
Colombia	204	34	16,67%
Mexico	692	100	14,45%

A traves de esta consulta podemos comprobar que las Skill con características de IA no son opcionales, sino fundamentales para poder ser tomados en cuenta en vacantes de DA, sobretodo si buscamos un puesto para una empresa NORTEAMERICANA

Modelo Relacional

```
USE JobMarketAnalysis;
GO
```

```
-- =====
-- LIMPIEZA PREVIA (si las tablas ya existen)
-- =====
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'FactJobPostings') DROP TABLE FactJobPostings;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'DimDate') DROP TABLE DimDate;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'DimCountry') DROP TABLE DimCountry;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'DimCompany') DROP TABLE DimCompany;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'DimIndustry') DROP TABLE DimIndustry;
IF EXISTS (SELECT * FROM sys.tables WHERE name = 'DimRemote') DROP TABLE DimRemote;
GO
```

```
-- =====
-- 1. TABLAS DE DIMENSIÓN
-- =====
```

-- 1.1 Dimensión Fecha

```
CREATE TABLE DimDate (
    DateKey INT PRIMARY KEY,
    DateValue DATE NOT NULL,
    Year INT,
    Quarter INT,
    Month INT,
    MonthName NVARCHAR(20),
    Weekday INT,
    WeekdayName NVARCHAR(20),
    IsWeekend BIT
);
```

```
INSERT INTO DimDate (DateKey, DateValue, Year, Quarter, Month, MonthName, Weekday, WeekdayName,
IsWeekend)
```

```
SELECT DISTINCT
    CAST(FORMAT(date_posted, 'yyyyMMdd') AS INT) as DateKey,
    date_posted as DateValue,
    YEAR(date_posted) as Year,
    DATEPART(QUARTER, date_posted) as Quarter,
    MONTH(date_posted) as Month,
    DATENAME(MONTH, date_posted) as MonthName,
    DATEPART(WEEKDAY, date_posted) as Weekday,
    DATENAME(WEEKDAY, date_posted) as WeekdayName,
    CASE WHEN DATEPART(WEEKDAY, date_posted) IN (1, 7) THEN 1 ELSE 0 END as IsWeekend
FROM vacantes_import
WHERE date_posted IS NOT NULL;
```

-- 1.2 Dimensión País

```
CREATE TABLE DimCountry (
    CountryKey INT IDENTITY(1,1) PRIMARY KEY,
    CountryName NVARCHAR(100) UNIQUE,
    Region NVARCHAR(50),
    Timezone NVARCHAR(50)
);
```

```

INSERT INTO DimCountry (CountryName, Region, Timezone)
SELECT DISTINCT
    pais_origen,
    CASE
        WHEN pais_origen IN ('USA', 'Canada') THEN 'North America'
        WHEN pais_origen IN ('Mexico', 'Colombia', 'Argentina', 'Chile', 'Peru', 'Uruguay', 'Costa Rica', 'Brazil') THEN
'Latin America'
        WHEN pais_origen = 'Spain' THEN 'Europe'
        ELSE 'Other'
    END as Region,
    NULL as Timezone
FROM vacantes_import
WHERE pais_origen IS NOT NULL;

```

-- 1.3 Dimensión Empresa (CORREGIDA - SIN DUPLICADOS)

```

CREATE TABLE DimCompany (
    CompanyKey INT IDENTITY(1,1) PRIMARY KEY,
    CompanyName NVARCHAR(500) UNIQUE,
    CompanyIndustry NVARCHAR(500)
);

```

```

INSERT INTO DimCompany (CompanyName, CompanyIndustry)
SELECT DISTINCT -- ← DISTINCT elimina duplicados como "Comcast"
    company,
    MAX(company_industry) as CompanyIndustry -- MAX para elegir un valor si hay varios
FROM vacantes_import
WHERE company IS NOT NULL AND company != ""
GROUP BY company; -- ← GROUP BY asegura unicidad

```

-- 1.4 Dimensión Industria

```

CREATE TABLE DimIndustry (
    IndustryKey INT IDENTITY(1,1) PRIMARY KEY,
    IndustryName NVARCHAR(100) UNIQUE,
    Keywords NVARCHAR(500)
);

```

```

INSERT INTO DimIndustry (IndustryName, Keywords) VALUES
('Tecnología/SaaS', 'tech,software,saas,startup,digital,cloud,ai,it'),
('Finanzas/Bancos', 'bank,financial,insurance,investment,fintech,seguro'),
('Salud/Farma', 'health,medical,pharma,biotech,hospital,clinical'),
('Retail/Ecommerce', 'retail,ecommerce,store,consumer,amazon,tienda'),
('Consultoría', 'consulting,advisory,big4,deloitte,accenture'),
('Manufactura', 'manufacturing,industrial,automotive,logistics'),
('Alimentos/Bebidas', 'food,aliment,bebida,restaurant,cocina,produccion'),
('Educación', 'education,university,e-learning,training'),
('Gobierno/ONG', 'government,public,non-profit,municipal');

```

-- 1.5 Dimensión Remoto

```

CREATE TABLE DimRemote (
    RemoteKey INT IDENTITY(1,1) PRIMARY KEY,
    RemoteType NVARCHAR(50) UNIQUE
);

```

```

INSERT INTO DimRemote (RemoteType) VALUES
('Remote'), ('Hybrid'), ('Onsite'), ('Not Specified');

```

-- =====

-- 2. TABLA DE HECHOS

```

-- =====

CREATE TABLE FactJobPostings (
    FactId INT IDENTITY(1,1) PRIMARY KEY,
    VacanteOriginalId INT,
    DateKey INT FOREIGN KEY REFERENCES DimDate(DateKey),
    CountryKey INT FOREIGN KEY REFERENCES DimCountry(CountryKey),
    CompanyKey INT FOREIGN KEY REFERENCES DimCompany(CompanyKey),
    IndustryKey INT FOREIGN KEY REFERENCES DimIndustry(IndustryKey),
    RemoteKey INT FOREIGN KEY REFERENCES DimRemote(RemoteKey),
    Title NVARCHAR(500),
    Description NVARCHAR(MAX),
    IsRemote BIT,
    Source NVARCHAR(100)
    -- termino_búsqueda eliminado (no existe en vacantes_import)
);

-- =====
-- 3. POBLAR TABLA DE HECHOS
-- =====

INSERT INTO FactJobPostings (
    VacanteOriginalId,
    DateKey,
    CountryKey,
    CompanyKey,
    IndustryKey,
    RemoteKey,
    Title,
    Description,
    IsRemote,
    Source
)
SELECT
    v.Id as VacanteOriginalId,
    CAST(FORMAT(v.date_posted, 'yyyyMMdd') AS INT) as DateKey,
    dc.CountryKey,
    dcomp.CompanyKey,
    NULL as IndustryKey, -- Se puede inferir después con otra consulta
    CASE
        WHEN v.is_remote = 1 THEN (SELECT RemoteKey FROM DimRemote WHERE RemoteType = 'Remote')
        WHEN v.description LIKE '%hybrid%' OR v.description LIKE '%hibrido%' THEN (SELECT RemoteKey
FROM DimRemote WHERE RemoteType = 'Hybrid')
        ELSE (SELECT RemoteKey FROM DimRemote WHERE RemoteType = 'Onsite')
    END as RemoteKey,
    v.title,
    v.description,
    v.is_remote,
    v.site as Source
FROM vacantes_import v
LEFT JOIN DimCountry dc ON v.pais_origen = dc.CountryName
LEFT JOIN DimCompany dcomp ON v.company = dcomp.CompanyName
WHERE v.date_posted IS NOT NULL;

-- =====
-- 4. VERIFICACIONES
-- =====

```

```
-- Verificar filas en cada tabla
SELECT 'DimDate' as Tabla, COUNT(*) as Filas FROM DimDate
UNION ALL
SELECT 'DimCountry', COUNT(*) FROM DimCountry
UNION ALL
SELECT 'DimCompany', COUNT(*) FROM DimCompany
UNION ALL
SELECT 'DimIndustry', COUNT(*) FROM DimIndustry
UNION ALL
SELECT 'DimRemote', COUNT(*) FROM DimRemote
UNION ALL
SELECT 'FactJobPostings', COUNT(*) FROM FactJobPostings;
```

```
-- Verificar integridad: buscar hechos sin país
SELECT COUNT(*) as HechosSinPais
FROM FactJobPostings f
LEFT JOIN DimCountry c ON f.CountryKey = c.CountryKey
WHERE c.CountryKey IS NULL;
```

```
-- Ver muestra de datos
SELECT TOP 10
    f.Title,
    c.CountryName,
    comp.CompanyName,
    r.RemoteType,
    f.IsRemote
FROM FactJobPostings f
JOIN DimCountry c ON f.CountryKey = c.CountryKey
JOIN DimCompany comp ON f.CompanyKey = comp.CompanyKey
JOIN DimRemote r ON f.RemoteKey = r.RemoteKey;
```



